1.3 Network devices

In this lesson

You will learn:

- more about network devices, software and rules.

Devices in a network

A network needs special devices. Each device does an important job in the network.

Servers

A **server** is a powerful computer. A network uses several servers. Each server does a different job.

Print server: When you print a picture or a document, the print server makes sure your file is printed correctly.

File server: This server makes sure your documents are saved correctly.

Email server: This server makes sure your emails reach the right people.



Switches and hubs

Switches are like major junctions in a road traffic system. When you send a message, a switch decides which cable the data needs to travel down to reach the right destination. For example, if you send an email, the switch makes sure your email goes to the email server. A **hub** is a type of switch.

Routers

A **router** connects a network to the internet. Your school network has a router. If you have an internet connection at home, it uses a small router.



How do network devices work together?

The devices in a network work together to get a job done. Here is what happens if you send an email from your school to your home.

You press 'Send'.

Your computer sends the email to a **switch**.

The switch sends the email to the **email server**.

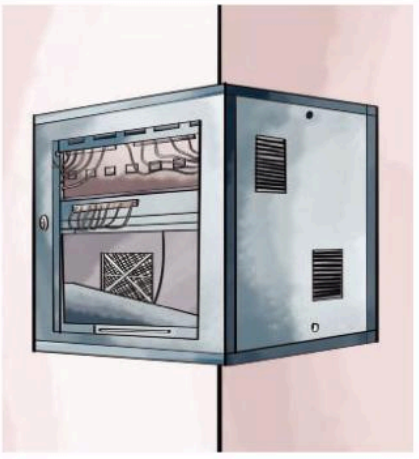
The email server sends the email to the **router**.

The router sends the email to the **internet**.

Your email reaches your home.

Where are network devices stored?

Network devices are usually stored in a locked room called a **server room**. They are stored in metal cabinets. Sometimes they are stored in small cabinets in classrooms.



Software and protocols

Two more things are needed to make a network work.

- Software is used to make sure the network runs properly. For example, network software is used to set up user logins.
- All parts of the network must understand each other. The whole network must follow rules so that jobs are done in the correct order. In a network the rules are called **protocols**.



Activity

Can you find any clues in your school that tell you the school has a network? If you can, take photos or make drawings as evidence.



Extra challenge

Read the description of what happens in a network when you send an email. Write a similar description of what happens when you print a document.



Explore more

If you have an internet connection at home, ask an adult to help you find the router. Be careful not to disturb any cables. Can you see anything that you recognise from studying this unit?

1.4 The internet

In this lesson

You will learn:

- what the internet is
- how the internet is different from the world wide web.

What is the internet?

The **internet** is a wide area network (WAN) that connects computers across the world. It is the biggest network in the world.

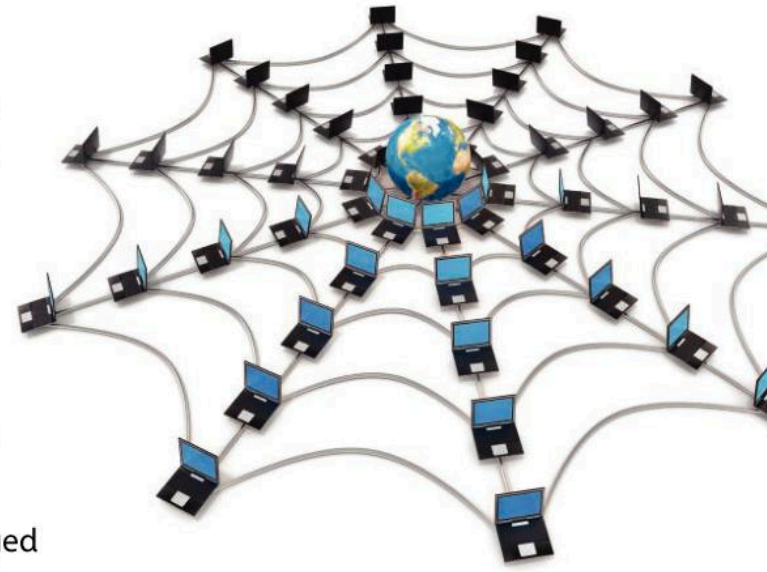
People use the internet to communicate and to share things with other people. The internet helps people to learn and work together.

The internet is not owned or controlled by a single person or organisation.

Internet services

The internet is a network of connections. Using those connections, we can make use of a range of services.

- **The world wide web:** Made of all the web pages in the world.
- **Email and chat:** Ways of sending messages and content to people you know.
- **Video conferencing:** People across the world can meet as if they were in the same room.
- **Streaming:** Games, music and videos can be played on your local device without being downloaded first.



The world wide web

The **world wide web** (www or just 'web') is a part of the internet. The web is all the websites and web pages stored on the internet. You use a web browser to look at web pages on the web. You use a search engine to search for information on the web.

The web is a powerful tool because every web page links to other pages. The links on web pages are called hyperlinks. If you drew lines between the hyperlinks on all the web pages on the web, it would look like a huge spider web.

Computers and the internet

You can use a computer to access all internet services. You use a browser to view the web. You can use other software applications (often called apps) for other internet services such as email.

Many people use a smartphone to access the internet. A smartphone can connect to the internet from anywhere that has a phone signal.

Smart devices

Many home devices are connected to the internet. These are called **smart devices**.

Home security systems can send a warning to your smartphone if sensors detect sound or movement in your home while you are out. You can check security cameras on your smartphone to see if there is an intruder.

You can use your smartphone to control home air conditioning and heating systems so the temperature is comfortable when you get home. You can switch on lights to make it look as if someone is home.





Activity

List the main internet services. Say which ones you have used. Give an example.



Extra challenge

Which devices do you use to connect to the internet? Which services do you prefer to access using a smartphone? Which services do you prefer to access using a computer? Give examples.

Think again

Research smart home heating or air conditioning systems on the web. List some of the features. How do smart home air conditioning and heating systems help the environment?